Bulls-Eye AI Learning Platform

6th Semester Group Project

1. Introduction

The **Bulls-Eye AI Learning Platform** is a smart educational web application developed as a **6th Semester Group Project**. The purpose of this platform is to enhance the learning experience by converting traditional study materials into **interactive quizzes using Artificial Intelligence**.

The system allows students to upload study materials such as PDFs and notes. Using AI technology, the platform automatically generates **Multiple Choice Questions (MCQs)** from the uploaded content. Students can then take real-time tests and receive instant results and performance analytics.

Currently, the project is in the **development phase**, and new features are continuously being added to improve functionality and user experience.

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2. Features

🗁 Upload Study Material

Students can upload study materials such as **PDF files and notes**. These materials are processed by the system to generate quizzes.

🤖 AI-Based MCQ Generation

The system uses **Artificial Intelligence** to automatically generate multiple-choice questions from the uploaded study materials.

📝 Real-Time Test System

Students can take quizzes and tests directly on the platform in real-time.

📊 Instant Result Evaluation

After completing a test, the system instantly evaluates the answers and displays the results.

📈 Performance Analytics

Students can track their performance through analytics, helping them understand their strengths and weaknesses.

🎨 Clean and Modern UI

The platform provides a modern and user-friendly interface to ensure a smooth learning experience.

📱 Fully Responsive Design

The system is fully responsive and works properly on **desktop, tablet, and mobile devices**.

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3. Technology Stack

Frontend

The frontend of the project is built using modern web technologies to create an interactive user interface.

- React.js
- Tailwind CSS
- JavaScript

Backend

The backend handles the server-side logic, API management, and data processing.

- Node.js

AI Integration

The system integrates Artificial Intelligence to generate quizzes and assist users through a chatbot.

- Ollama AI

Database Integration

A database system is used to store user data, quizzes, and study materials.

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4. AI Chatbot

The platform includes an **AI-powered chatbot** that helps students with their queries.

The chatbot is powered by **Ollama AI** and can assist students with:

- Questions related to study material
- Understanding quiz topics
- General learning assistance

This feature improves the overall learning experience by providing instant support to students.

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5. Project Objective

The main objective of this project is to **transform static study material into interactive quizzes using Artificial Intelligence**.

The platform helps students to:

- Improve self-assessment
- Practice topic-wise MCQs
- Save preparation time

- Track their learning performance efficiently

This system aims to make learning **more interactive, intelligent, and efficient**.

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6. How to Run the Project

Follow the steps below to run the project on a local machine.

Prerequisites

Make sure the following software is installed on your system:

- Node.js (v14 or higher)
- npm or yarn
- Git for version control
- Ollama AI (optional for chatbot functionality)

Installation & Setup

1. Clone the Repository

```
git clone <repository-link>
cd Bulls-Eye-Mern--Project
```

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2. Install Backend Dependencies

```
cd backend
npm install
```

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3. Install Frontend Dependencies

```
cd ../AI-Learn
npm install
```

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7. Running the Application

Start Backend Server

```
cd backend
node server.js
```

Backend will run on:

http://localhost:5000

Start Frontend Development Server

```
cd AI-Learn
npm run dev
```

Frontend will run on:

http://localhost:5173

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8. Build for Production

To build the frontend for production:

```
cd AI-Learn
npm run build
```

This will generate optimized production files.

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9. Future Enhancements

In future versions, the following features can be added:

- User authentication system
- Admin dashboard
- AI-powered question difficulty levels
- Topic-wise performance tracking
- Leaderboard system for students
- Cloud database integration

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10. Conclusion

The **Bulls-Eye AI Learning Platform** is designed to modernize the traditional study process by integrating **Artificial Intelligence and interactive testing systems**.

By transforming static study materials into dynamic quizzes, the platform helps students improve their preparation, track their progress, and enhance their learning efficiency.

This project demonstrates how AI can be effectively used in the **education sector** to create smart learning solutions.

Bulls-Eye Project – Viva Questions with Answers

1. What is the title of your project?

Answer:
The title of our project is **"Bulls-Eye AI Learning Platform."** It is an AI-based system that converts study materials into interactive quizzes.

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2. What is the objective of your project?

Answer:
The main objective of our project is to **convert static study material into interactive MCQ quizzes using Artificial Intelligence** to improve students' learning and self-assessment.

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3. Why did you choose this project?

Answer:
We chose this project because many students struggle with self-evaluation while studying. Our platform helps students **practice quizzes and track their performance automatically**.

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4. What technologies are used in your project?

Answer:
The technologies used in our project are:

Frontend:

- React.js
- Tailwind CSS
- JavaScript

Backend:

- Node.js

AI Integration:

- Ollama AI

Database:

- Database integration for storing user data and quizzes.

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5. What is React.js?

Answer:
React.js is a **JavaScript library used for building interactive user interfaces**, especially for single-page web applications.

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6. Why did you use React.js?

Answer:
We used React.js because it provides **fast rendering, reusable components, and better performance** for modern web applications.

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7. What is Node.js?

Answer:
Node.js is a **JavaScript runtime environment that allows developers to run JavaScript on the server side**.

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8. What is the role of the backend in your project?

Answer:
The backend handles **server operations, API requests, AI processing, and database communication**.

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9. What is Artificial Intelligence in your project?

Answer:
Artificial Intelligence is used to **analyze study materials and automatically generate multiple-choice questions (MCQs)**.

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10. What is Ollama AI?

Answer:
Ollama AI is an **AI framework that allows running large language models locally**. In our project, it is used to provide **chatbot support for students**.

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11. What is the function of the AI chatbot?

Answer:
The AI chatbot helps students by **answering questions related to study material and quizzes**.

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12. How does the MCQ generation work?

Answer:
The system analyzes uploaded study materials using AI models and **extracts key information to generate multiple-choice questions automatically**.

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13. What is the benefit of this system?

Answer:
The system helps students to:

- Practice MCQs
- Evaluate their knowledge

- Track performance
- Save preparation time

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14. What are the main features of your project?

Answer:

Main features include:

- Upload Study Material
- AI-based MCQ generation
- Real-time tests
- Instant result evaluation
- Performance analytics
- AI chatbot support

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15. What is a responsive design?

Answer:
Responsive design means that the website **automatically adjusts its layout for different screen sizes like mobile, tablet, and desktop**.

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16. What are the limitations of your project?

Answer:
Some limitations are:

- AI accuracy depends on the quality of study material
- Requires internet connection
- AI processing may take time for large documents

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17. What improvements can be added in the future?

Answer:
Future improvements may include:

- User authentication system
- Admin dashboard
- Leaderboard system
- Mobile application
- Cloud-based AI processing

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18. What is the role of Tailwind CSS?

Answer:
Tailwind CSS is used to **design the user interface quickly using utility-based CSS classes**.

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19. What type of project is this?

Answer:
This is a **Full Stack AI-based Web Application**.

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20. What is the conclusion of your project?

Answer:
Our project demonstrates how **Artificial Intelligence can improve the learning process** by transforming traditional study materials into interactive quizzes and helping students track their progress effectively.